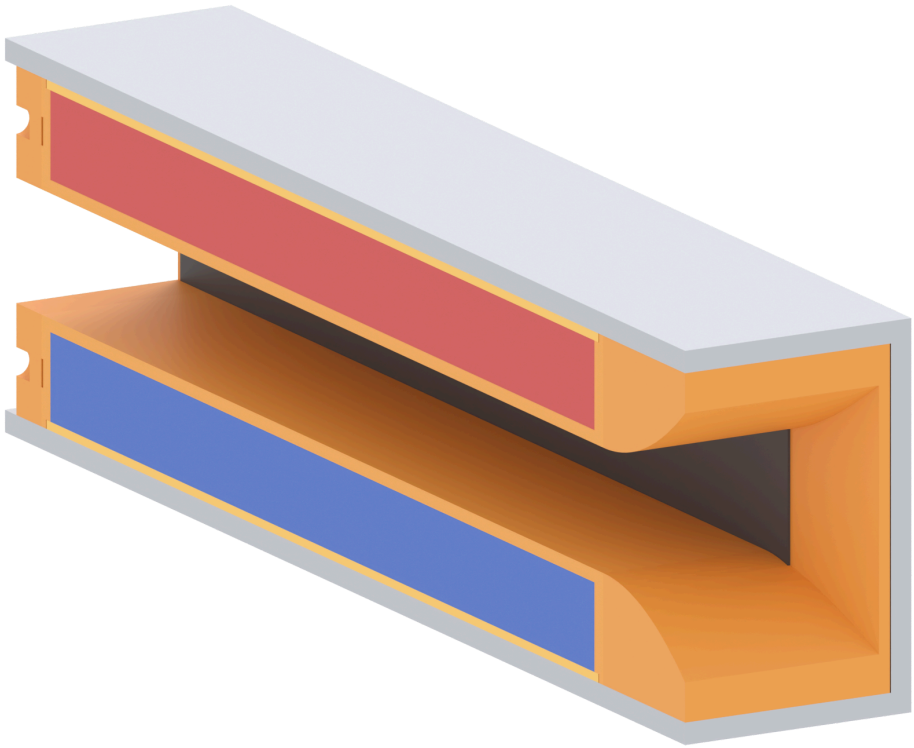
Magnetide v21-r2g — Manufacturing & Assembly

Reynard Filbert Tang · Kelas 10.8, Kelompok 1 · MHD seawater pump
Envelope 150 × 72 × 44 mm · bore 40 × 14 × 124 mm · all dimensions in mm
Every figure is a render of the delivered solid, not a drawing of it.

What you are building

Brine runs through a 40 × 14 mm bore. A 483 mT field crosses it, current flows between two graphite plates on the side walls, and the Lorentz force pushes the water out the straight end. The yoke is the magnetic return path and the tray is the only thing keeping water away from the magnets — both get checked automatically every time the model rebuilds.



Cut on the centre plane. Bore at the bottom, 1.5 mm tray floor above it, magnet in the closed cavity, 1 mm cap between magnet and steel plate. The floor is continuous over the electrode groove.

Stack, per half (z from the mid-plane)

z	Layer	Thickness	Function
0 – 7	bore	14 tall	water channel, 40 wide
7 – 8.5	tray floor	1.5	the only barrier between water and magnets
8.5 – 18.5	magnet cavity	10	closed on all six sides
18.5 – 19.5	cap	1.0	locks the stack down, flush with the steel
19.5 – 22	steel plate	2.5	flux return

Parts

Metal — `metal/`

File	Qty	Material	Notes
`clamshell-flat-2.5mm.dxf`	2	2.5 mm mild steel (DC01 / SPCC / AISI 1008)	Yoke blank **107.83 × 150**. Plate plus two side walls, and the split-line flange cropped into **four lugs of 16 mm** per side — the lug pattern, not a continuous band. 317 g per half.

Printed — `plastic/`

File	Qty	Size	Notes
`tray.stl`	2	124 × 50.7 × 19.5	Closed magnet cavity, 1 mm cap ledge, 1 mm electrode groove. Floor down.
`cap.stl`	2	121.7 × 42.7 × 1.0	Thin lid, 100 % infill — a clamp, not a shell.
`bellmouth.stl`	1	20 × 49.7 × 39	48 × 30 mouth to 40 × 14 over 20 mm.
`rear_block.stl`	1	5.5 × 49.7 × 39	Outlet, 5 mm thick. Carries the lead channel.

Bought

Item	Qty	Notes
N52 tile 20 × 10 × 5, nickel plated	96	48 per face: 6 along × 4 across × 2 deep.
Graphite plate, 134 × 14 × 2	2	124 mm wetted; ends sit 5 mm inside the end blocks.
M3 × 16 bolt + nut + washer	8	Through the yoke lugs at the split line.
Epoxy (30 min+)	—	Magnets to tray, and the lead exit after wiring.

Manufacturing

Steel: laser + 4 folds

Parameter	Value	Where it comes from
Sheet thickness	2.5	one gauge, plate and wall
Flat blank	107.83 × 150	56 plate + 2 × (19.96 wall + 5.96 lug)
Fold 1 — wall	at ±28	down 90°, forms the bore side
Fold 2 — lug flange	at ±47.96	out 90°, lugs only, 4 × 16 mm per side
Bend allowance	2.042	K = 0.44, R_{in} = 0.2
Mass per half	317 g	7.85 g/cm³

> **Cut one test blank first.** K = 0.44 is nominal. At K = 0.40 the lug holes drift 0.3 mm and the two halves stop lining up — measure the folded part before cutting the second blank.

Printing

Part	Layer	Walls	Infill	Orientation
tray	0.2	4	40 %	floor down, cavity open-side up, no supports
cap	0.16	3	100 %	flat
bellmouth	0.2	3	25 %	mouth up
rear_block	0.2	4	40 %	flat face down

Watertight: the three paths that were open

The magnet cavity had three ways in. All three are closed, and each one is a hard assert in the build script — the model refuses to export if any reopens.

Leak path	What was wrong	Fix
electrode groove roof	the 2.2 mm groove was cut 3.0 deep from z 7.0, so its ceiling sat at **z 10.0 — 1.5 mm above the cavity floor at 8.5**. The groove ran the full 124 mm with an open top: a gutter into the magnet pocket.	groove cut **1.0 deep** (7.0 → 8.0), ceiling 0.5 mm below the cavity floor. 0.5 mm of floor covers the electrode lip, 4.85 mm of solid wall sits behind it.
end-block peg holes	pegs were located with holes drilled 2.6 mm into a 2.0 mm end wall — both ends were **through-holes into the cavity**.	blind holes **1.6 deep**, 0.4 mm of plastic left behind, plus a 2.3 mm lead-in. Pegs shortened to 2.0 mm.
magnet lead exit	a full-length bore through the rear block opened the cavity to the outside.	blind channel from the outside of the outlet wall, stopping **0.4 mm short**. You drill it through at build time and epoxy around the wire.

> **One joint is still plastic-on-plastic:** the two trays butt at z 0 with no gasket, because that is the v21-r2e arrangement. It is a face-to-face butt joint at the bore ceiling, so it leaks at most a weep. If this pump ever runs pressurised, put a 1 mm gasket there — it costs 1 mm of lug gap and some field.

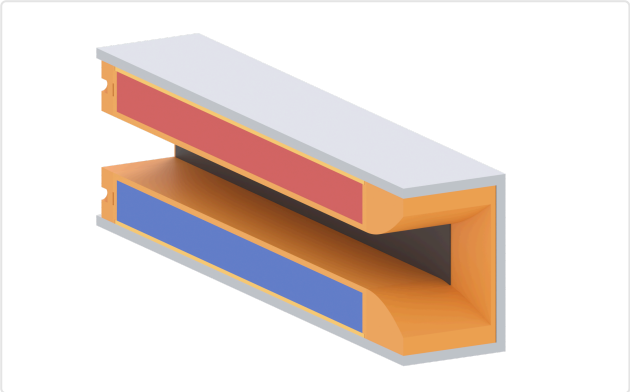
Build sequence

1 Fold the two yokes

Cut two blanks. Fold 90° down at ±28 (the walls), then fold the lugs out 90° at ±47.96. Four lugs per side, 16 mm long, one Ø3.4 hole each.

Check: 56 mm across the plate, 22 mm plate face to lug face, holes on the split line at ±50.94.

2 Stack the magnets into the tray cavity



Eight tiles per row, four rows, two layers deep — 48 per face. Two 5 mm layers make the 10 mm bar and fill the cavity 8.5 → 18.5.

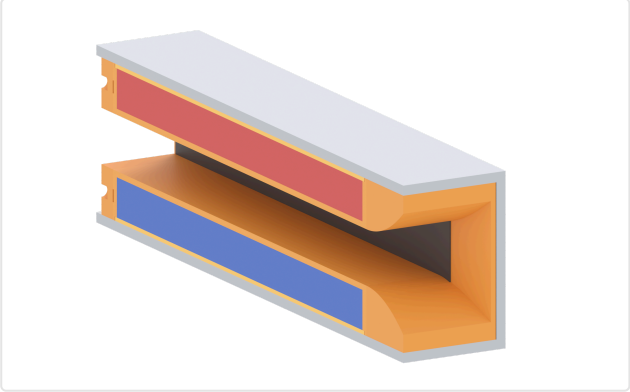
Compass-check every row before the epoxy sets. All four rows on one face point the same way; the opposite face points the other way.

> Tiles are modelled as 9.7 mm bars with 0.3 mm clearance so they drop in. Do not press-fit — a cracked tile corrodes from the crack.

3 Cap it

Drop the 1 mm cap into the ledge in the tray rim. It sits on the magnets, top face flush with the steel inner face, and stops the stack lifting when the field pulls. A bead of epoxy at the rim, never a puddle — epoxy in the bore is unreachable once closed.

4 Seat the electrodes



Each 134 × 14 × 2 plate slides into the 1 mm groove from the end. Its top shoulder hides behind 0.5 mm of floor; the full 14 mm face is exposed — 124 × 14 = **1736 mm²** per electrode.

Push each plate until its end sits 5 mm inside the bellmouth or rear-block pocket. That pocket is the end seal.

5 Bellmouth and rear block

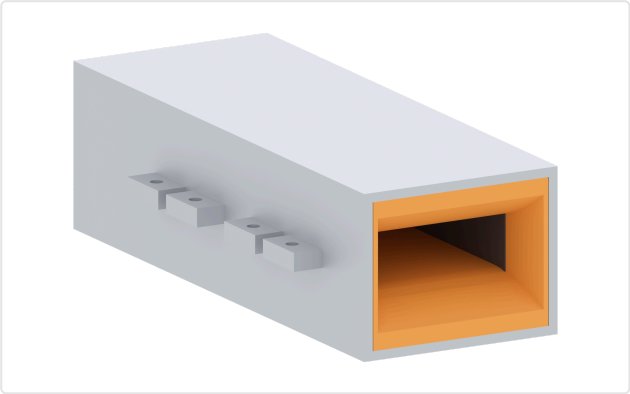
Two 2 mm pegs each, into the blind holes in the tray end faces. The bellmouth takes the 48 × 30 mouth to the bore over 20 mm.

At the outlet, drill through the last 0.4 mm of the lead channel, feed the two wires out, epoxy the hole closed around them. Keep the outlet straight — a diverging nozzle measured 0.3 → 0.2 m/s.

6 Close the yoke

Lower tray into the bottom yoke, upper into the top, bring the halves together so the lug faces meet. Eight M3 through the lugs, tightened in a cross pattern a little at a time.

The lugs are 8 mm strips around Ø3.4 holes, so location comes from the bolt shanks. Add two 3 mm dowels through the lug pairs if it will get knocked around.



Fit and clearance

Joint	Clearance	Consequence if wrong
magnet in cavity	0.3 per side	press fit cracks tiles
cap in rim ledge	0.15 per side	cap lifts, magnets float up
tray in yoke throat	0.15 per side	tray rattles, wears through the 1.5 mm floor
electrode in groove	1.0 deep, 0.2 wide	too deep = groove opens into the cavity
electrode end in pocket	5 mm deep	short end = leak behind the electrode
peg in end wall	1.6 blind	deeper = hole through into the magnets
M3 in lug hole	Ø3.4	standard M3 clearance

Before you power it

- ☐ Both halves flush, 8 bolts seated in the lugs, no gap at the lug faces
- ☐ Compass: every row on a face points the same way, the two faces oppose
- ☐ Tray floor unbroken over the whole magnet area — hold it up to a light
- ☐ Electrodes fully home in their end pockets, not shorting to steel through a scratch
- ☐ Lead exit drilled, wired and epoxied; bore clear down its whole 124 mm
- ☐ Fuse in the supply line; runs under 2 minutes, ventilated (Cl_2)

> **Model status.** Dimensions read back off the delivered solid. Magnets modelled as four 120 × 9.7 × 10 bars per face instead of 96 loose tiles: same envelope, same 40 mm coverage, readable in a render.

Magnetide v21-r2g · generated from `Magnetide-v21r2g-assembly.step` · 18 solids · bore verified clear 40 × 14 × 124 · yoke blank 107.83 × 150 × 2.5 · probes: groove roof, peg holes, lead exit, bore